

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 2C

Application: investigate evidence and design a better study



Name _____ Class _____ Date _____

A fictional college has **3,696 students**. The canteen may add a smaller portion alongside its usual portion. The report claims **about 68% support** and **about 26% less food waste**.

Calculator allowed. Show your working and round percentages to 2 decimal places.

EXERCISE C1.1 The evidence behind the waste claim

DISCUSS 2 MIN

Independent pilot: the same menu on two days, using an accurate, checked scale.

Portion	Mass served	Meals	Total leftovers
Usual	425 g	173	14,186 g
Smaller	315 g	173	15,397 g

a) Calculate the percentage change in portion size and total leftovers.

Portion size: working and percentage change

Leftovers: working and percentage change

b) What does the pilot establish about the waste claim? Explain why the two percentage changes differ.

EXERCISE C1.2 What can the missing replies change?

DISCUSS 3 MIN

237 environmental-club members were invited. **143 replied**, and **97 of those respondents supported** the option.

a) Calculate both rates. b) Find the lowest and highest possible support among all 237 invitees.

Response rate

Support among respondents

Lowest possible support among all 237 invitees

Highest possible support among all 237 invitees

Can we establish a majority among the invitees? Explain.

REPAIR THE INVESTIGATION

EXERCISE C2.1 What needs repairing? DISCUSS 2 MIN

The original investigation had these further problems:

- 1. Every invitee belonged to the environmental club.
- 2. The question asked: "Don't you agree that smaller portions are an excellent way to protect the environment?"
- 3. Its scale recorded leftover masses **12 g too high**.

The independent pilot in C1.1 used a different, accurate scale.

Name each problem, identify the claim or measurement it threatens, and specify a repair.

Fault	Name the problem	What does it threaten?	A practical repair
Club-only invitations			
Question wording			
Faulty scale			

EXERCISE C2.2 A question that tests the proposal DISCUSS 2 MIN

The actual proposal is to **add a 315 g option** while keeping the **usual 425 g portion** available.

Write one neutral survey question.

Give balanced response options.

Pair check: Ask another pair to explain what your question means. Did they understand the same proposal? Record one useful comment or change.

CHOOSE A DEFENSIBLE SELECTION PROCESS**The sampling brief**

Business	Engineering	Total students	Target invitations
2,173	1,523	3,696	176

There are **84 classes of 44 students**. Each student belongs to one class, and classes may mix programmes. Complete student, programme and class lists are available.

Goal: College-wide opinions and separate findings for each programme.

EXERCISE C3.1**Two ways to draw 176 students****DISCUSS 3 MIN**

Procedure A	Procedure B
Uniformly select 176 different IDs from 1 to 3,696.	Form 1,848 fixed pairs. Randomly select 88 different pairs and invite both students in each. Example pairs: 1 with 1,849; 2 with 1,850.

a) What fraction of students does each procedure select?

b) Which is a simple random sample of individual students? Explain.

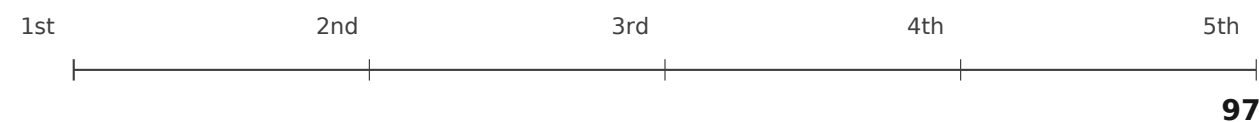
Describe a group that Procedure B cannot select.

EXERCISE C3.2**A systematic selection record**

Population: 3,696. Target: 176. The fifth selected ID is 97.

a) Find the interval and starting ID. Complete the first five selections.

Sampling interval: working	Starting ID: working
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b) Can these IDs prove that the start was chosen randomly? What would you need to check?

FROM INVITATIONS TO REPLIES

EXERCISE C3.3 Is the proposed split closest?

DISCUSS 3 MIN

Allocate **176 invitations** proportionally between the programmes.

Programme	Population	Proposed invitations
Business	2,173	104
Engineering	1,523	72
Total	3,696	176

Business: ideal allocation and rounding

Engineering: ideal allocation and rounding

Is the split closest? Explain your rounding and how to select within each programme.

EXERCISE C3.4 What if we sample whole classes?

DISCUSS 3 MIN

An alternative randomly selects **four of the 84 classes**, then invites every student in those classes.

Programme	College	Selected sample
Business	2,173	147
Engineering	1,523	29

a) Compare Engineering's percentage in the college and selected sample.

Engineering in the college: working and %

Engineering in the sample: working and %

b) Which plan best supports separate programme findings next time? Does this draw prove a biased procedure? Explain.

EXERCISE C3.5 Who replied to the repaired survey?

DISCUSS 3 MIN

The team uses the stratified plan. These questionnaires come back:

Programme	Invited	Replied
Business	103	84
Engineering	73	24
Total	176	108

Business response rate

Engineering response rate

Engineering share of respondents

Propose a follow-up based on the pattern you find.

RECORD THE DATA AND WRITE THE REPORT

EXERCISE C4.1 The canteen data record

Classify each variable as **categorical** or **quantitative**. If quantitative, state **discrete** or **continuous**.

Item	Recorded variable	Classification
A	Student ID: 03752	
B	Programme code: 1 Business / 2 Engineering	
C	Meals bought in seven days: 8	
D	Portion mass rounded to the nearest gram: 315 g	
E	Exact waiting time, recorded as 7.3 minutes	
F	Waiting-time band: under 5 / 5-10 / over 10 min	
G	Opinion code: 1 Support / 2 Oppose / 3 Unsure	
H	Leftover mass rounded to the nearest gram: 84 g	

Explain C (meals bought).

Explain D (rounded portion mass).

Explain G (opinion code).

Which variable directly measures food waste?

EXERCISE C4.2 The college needs a corrected report

DISCUSS 3 MIN

Write **three sentences, up to 75 words**. Use the evidence already calculated.

1. State what the original support figures establish.
2. State what the pilot establishes about portions and waste.
3. Explain what further evidence or action the college needs.

SEMINAR DISCUSSIONS 1-3

All tables discuss the same question. Write brief notes, then check the answer in class.

EXERCISE C5.1 Whose population?

DISCUSS 2 MIN

Warehouse A inspects all **60,000 units** in its current stock. **5,000 are defective: 8.33%**. The company has **eight warehouses**.

Research concerns	Inspected units: population or sample?	The 8.33%: parameter or statistic?
Warehouse A's current stock		
Current stock across all eight warehouses		

Why can the same records have different labels?

EXERCISE C5.2 A cleaner streaming dataset?

DISCUSS 3 MIN

A streaming survey randomly selects **157 students**. **117 reply**, including **19 who report zero spending**. The analyst removes those 19 zeros and adds **40 volunteers** to replace the students who did not reply.

1. How many records are in the final dataset? Show your working.

2-3. Identify two problems and explain how to repair the process.

Change	Problem	Repair
Remove the 19 zeros		
Add 40 volunteers		

EXERCISE C5.3 Equal numbers, equal chances?

DISCUSS 3 MIN

The city randomly selects **50 residents from each age group**.

Age (completed years)	Population	Selected	Selection chance: working and %
0-18	1,846	50	
19-40	3,927	50	
41 and above	2,615	50	

a) Is this stratified? Is it proportional? Explain.

b) Compare the highest and lowest selection chances using a ratio.

SEMINAR DISCUSSIONS 4-6

EXERCISE C5.4 Who is at home?

DISCUSS 2 MIN

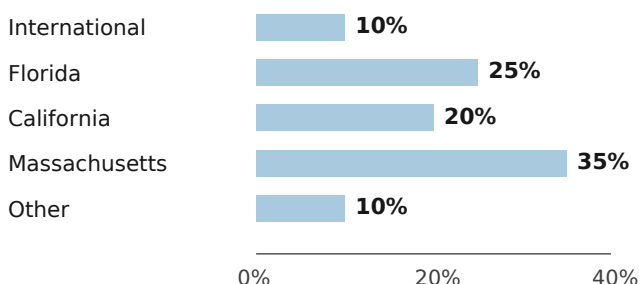
A health survey randomly selects **10 neighbourhoods**. Interviewers attempt every household on the complete lists, but visit only **9-11 am on Tuesday** and make **no return visits**.

Discuss	Your notes
Who might be missed?	
When could this distort health findings?	
Can you predict the direction?	
Give one practical repair.	

EXERCISE C5.5 Which headline is justified?

DISCUSS 2 MIN

120 Boston residents report their holiday preferences.



A. "35% of Boston residents prefer Massachusetts."

B. "Massachusetts is the top choice among the 120 respondents."

C. "The sample is representative because the percentages total 100."

Choose a supported headline. Rewrite one unsupported headline.

EXERCISE C5.6 A pattern in the register

DISCUSS 3 MIN

Each block of 21 register positions has **20 day students**, then **one evening student**. Select every 21st student.

Block	Day students	Evening student
1	Positions 1-20	Position 21
2	Positions 22-41	Position 42
3	Positions 43-62	Position 63

a) What happens with each random start?

Random start	First four selected positions	Who enters?
13		
21		

Does a random start solve this pattern? Suggest a change.

SEMINAR DISCUSSION 7 AND INDIVIDUAL EXIT

EXERCISE C5.7

A method that fits the job

DISCUSS 3 MIN

Choose a method and one clear selection step for **each** case. Make your procedure meet the stated requirements.

A. Online university survey

A complete contact list is available. Select **157 of 4,286 students**.

Method and selection step:

B. Programme comparison

There are **3,287 students**, including **148 Engineers**. Of **173 invitations**, at least **40** must go to Engineers.

Method and selection step:

C. In-person health study

There are **137 villages**. The team can visit **six villages** and contact every household there.

Method and selection step:

EXERCISE C6

What can this report tell you?

INDIVIDUAL 4 MIN

Another college numbers its student register from **1 to N**.

"We selected every 39th student. The fifth selected ID was 184."

a) Find the starting ID. Show your reasoning.

b) Can you determine the exact sample size from this report? Explain.

c) Give two possible population sizes consistent with the report that would produce different sample sizes.

Possible population N	Selected sample size	Last selected ID